

Power Electronics — Chapter 1

DC-DC Converters: Buck Converter

POWER_E_466

Buck Converter

The buck converter is a DC-DC converter that steps down voltage ($V_{out} < V_{in}$).

Motivating the topology

Start from a black-box model: given $V_{in} = 50\text{ V}$, we want an average output of $V_{out} = 20\text{ V}$ across a resistive load.

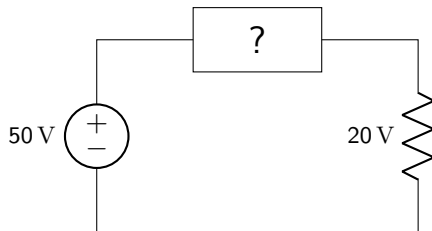


Figure: Black-box step-down converter: $V_{in} = 50\text{ V}$ in, unknown network, $\langle V \rangle = 20\text{ V}$ across the load.

Ideal switch model

The simplest way to reduce an average voltage is to switch the input on and off periodically (an ideal SPDT switch connecting the load either to V_{in} or to ground), so that the average value is controlled by the fraction of time spent connected to V_{in} — the *duty cycle* D :

$$\langle V \rangle = D V_{in}.$$

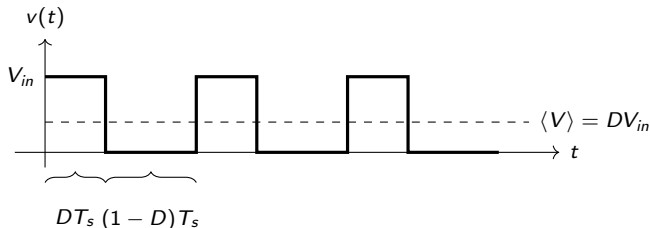


Figure: Ideal switch model: pole connects to V_{in} for a fraction D of the switching period T_s , and to ground for the remaining $(1 - D)T_s$, producing a PWM waveform whose average is $\langle V \rangle$.

Why we need an LC filter

The switched (PWM) waveform is not a clean DC voltage — it still contains switching-frequency ripple. From a Fourier-series point of view, the desired DC output corresponds only to the $f = 0$ Hz (DC offset) term; every other harmonic should be attenuated. A low-pass *LC filter* does exactly this: it passes the DC component while suppressing the switching-frequency content.

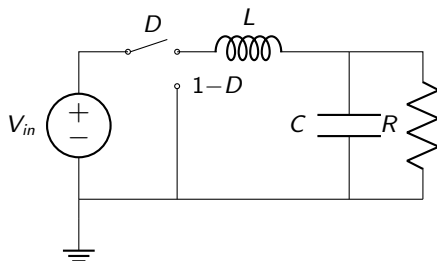


Figure: Buck converter: switch D followed by an LC low-pass filter, driving a resistive load. This is the buck converter topology.

Inductor and capacitor averaging identities

For an inductor,

$$v = L \frac{di}{dt} \iff i = \frac{1}{L} \int v dt,$$

so if i is periodic in steady state, the average inductor voltage must be zero:

$$\langle v_L \rangle = 0 \quad (\text{volt-second balance}).$$

For a capacitor,

$$i = C \frac{dv}{dt} \iff v = \frac{1}{C} \int i dt,$$

so if v is periodic in steady state, the average capacitor current must be zero:

$$\langle i_C \rangle = 0 \quad (\text{current-second/charge balance}).$$

Steady-state inductor current ripple

Over one switching period T_s , the inductor current i_L ramps up while the switch connects to V_{in} (duration DT_s) and ramps down while it connects to ground (duration $(1 - D)T_s$):

$$\frac{di_L}{dt} = \frac{v_L}{L}, \quad 2\Delta i_L = \frac{V}{L} DT_s \quad (\text{rising}), \quad 2\Delta i_L = \frac{V}{L} (1-D)T_s \quad (\text{falling})$$

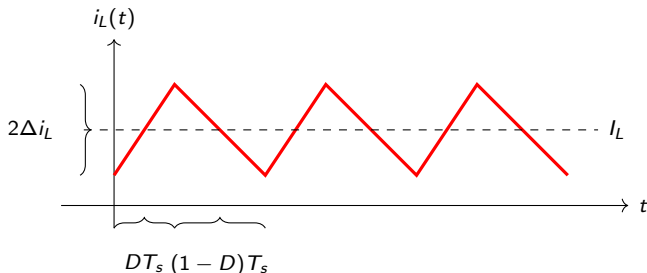


Figure: Steady-state inductor current: triangular ripple of peak-to-peak $2\Delta i_L$, rising for DT_s and falling for $(1 - D)T_s$ each switching period.

Worked example: switch ON

Given $V_{in} = 50\text{ V}$, $D = 0.5$, $L = 10\text{ mH}$, $T_s = 1\text{ ms}$, and (for this simplified example) an output branch modelled as a current source with $V = 0$:

$$\frac{di_L}{dt} = \frac{50}{10^{-3}} = 50 \times 10^3 \text{ A/s} \rightarrow \Delta i_L = 5 \text{ A}.$$

At $t = 0^+$: $i_L = 5\text{ A}$, $i_O = 0$, $i_C = i_L - i_O = 5\text{ A}$.

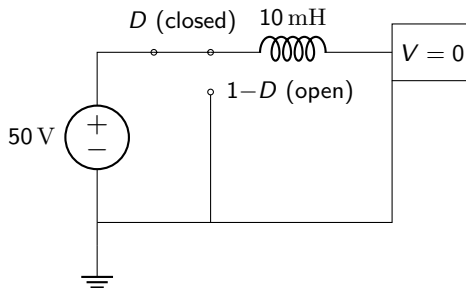


Figure: Buck circuit during the ON interval: $V_{in} = 50\text{ V}$, $D = 0.5$, $L = 10\text{ mH}$, $T_s = 1\text{ ms}$, output branch at $V = 0$.

Worked example: switch OFF

$$\frac{di_L}{dt} = \frac{0}{10^{-3}} = 0 \text{ A/s},$$

so i_L stays constant at 5 A while the switch is off (for this particular example, since the inductor sees 0 V in both states here).

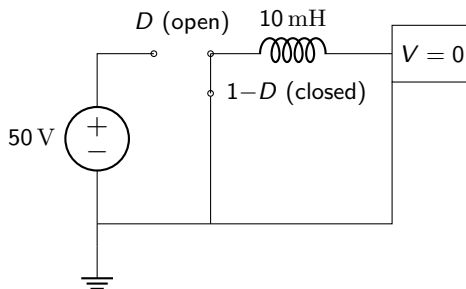


Figure: Buck circuit during the OFF interval, same component values as above.

Is this steady state?

Not necessarily — V_{out} can still change from cycle to cycle if the circuit is not yet in continuous, periodic operation. As i_O increases from 0, $i_C = i_L - i_O$ decreases; by charge balance, $\langle i_C \rangle = 0$ is exactly the condition that defines the steady-state operating point.

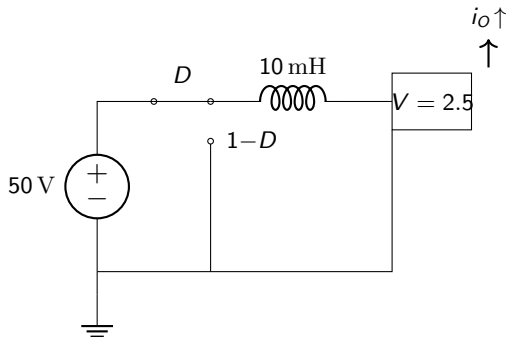


Figure: Buck circuit with increased output current ($V \approx 2.5\text{ V}$ example): as $i_O \uparrow$, $i_C \downarrow$, illustrating the approach to $\langle i_C \rangle = 0$ (charge balance).

Approaching steady state

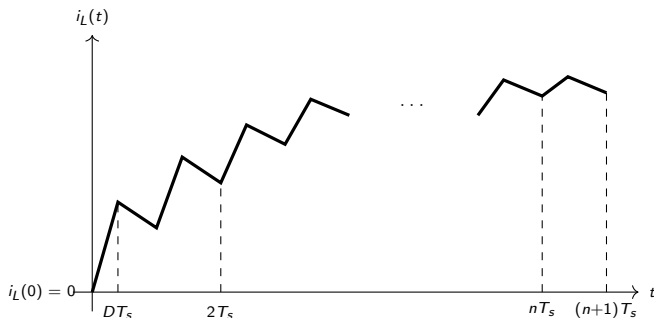


Figure: Classic multi-period inductor current waveform $i_L(t)$ ramping toward steady state: slope $\frac{V_g - v(t)}{L}$ while on, slope $\frac{-v(t)}{L}$ while off, converging so that $i_C \approx 0$ in steady state (cf. Erickson, *Fundamentals of Power Electronics*).

Realizing the switch with real devices

The ideal SPDT switch (poles A , B , C) is not a physical component. It is realized using one *active* switch (a transistor, controlled by a gate/base signal) for the A – B throw, and one *passive* switch (a diode, which conducts automatically depending on the circuit's own voltage/current) for the C – B throw.

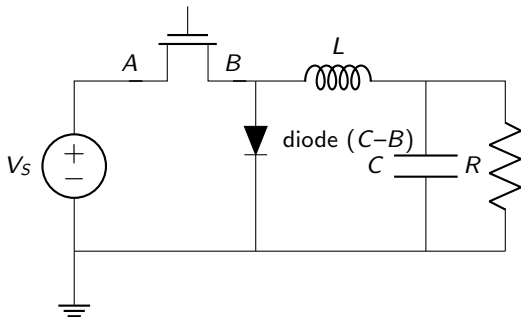


Figure: Replacing the ideal switch with a MOSFET (A – B) and a diode (C – B), giving the practical buck converter realization.

Switch analysis: node A–B (transistor position)

When the ideal switch is ON: $V_{AB} = 0$, $I_{AB} = I_L$. When OFF: $V_{AB} = V_S$, $I_{AB} = 0$. This (I, V) locus (first quadrant only: on along the current axis, off along the voltage axis) matches an *active* switch such as a MOSFET/BJT/IGBT.

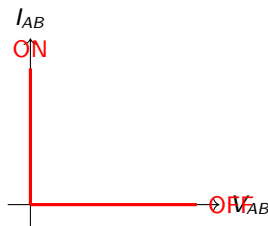


Figure: I_{AB} – V_{AB} switch locus for the transistor position: ON along the current axis, OFF along the voltage axis.

Switch analysis: node C–B (diode position)

When the ideal switch is ON (meaning the A – B throw is closed):
 $V_{CB} = -V_S$, $I_{CB} = 0$. When OFF (meaning C – B conducts instead):
 $V_{CB} = 0$, $I_{CB} = I_L$. This locus (blocks negative voltage, conducts positive current, no external control terminal) matches a *passive* switch: a diode.

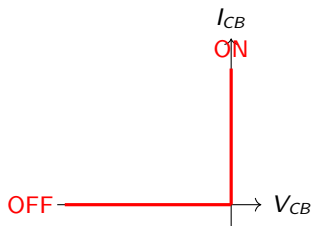


Figure: I_{CB} – V_{CB} switch locus for the diode position: blocks reverse voltage, conducts forward current.

Switch device characteristics

- **BJT / IGBT** are active switches: the conducting state is set entirely by the control-terminal signal, independent of the waveforms at the power terminals. Ideal characteristic: $i = 0$ when off (blocks $v \geq 0$); $v \approx 0$ when on (conducts $i \geq 0$). Poor/no reverse-conducting or reverse-blocking capability.
- **MOSFET** behaves similarly, but can additionally conduct in reverse. It is normally operated with $i \geq 0$, same as BJT/IGBT, so an active SPST switch can be realized with a BJT, IGBT, or MOSFET.
- **Diode** is a passive switch with no control terminal: its state is determined entirely by the terminal waveforms $i(t)$, $v(t)$. Ideal characteristic: off ($i = 0$) when $v < 0$; on ($v = 0$) when $i > 0$. It can block negative voltage but not positive voltage.

Switch device characteristics (reference figures)

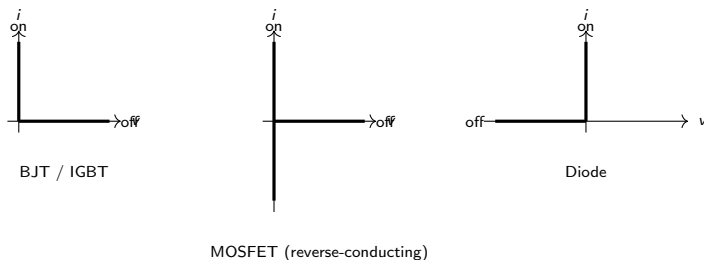


Figure: Idealized switch characteristics for BJT/IGBT, MOSFET, and diode (simplified from the textbook reference figures — the real-world I - V curve families are omitted here).

Buck: Summary

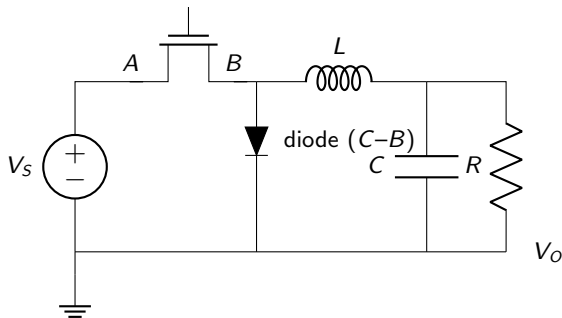


Figure: Final buck converter circuit: MOSFET (A-B), freewheeling diode (C-B), LC output filter, load.

$$\boxed{V_O = D V_S} \quad \text{with } \langle i_C \rangle \rightarrow 0 \text{ in steady state.}$$